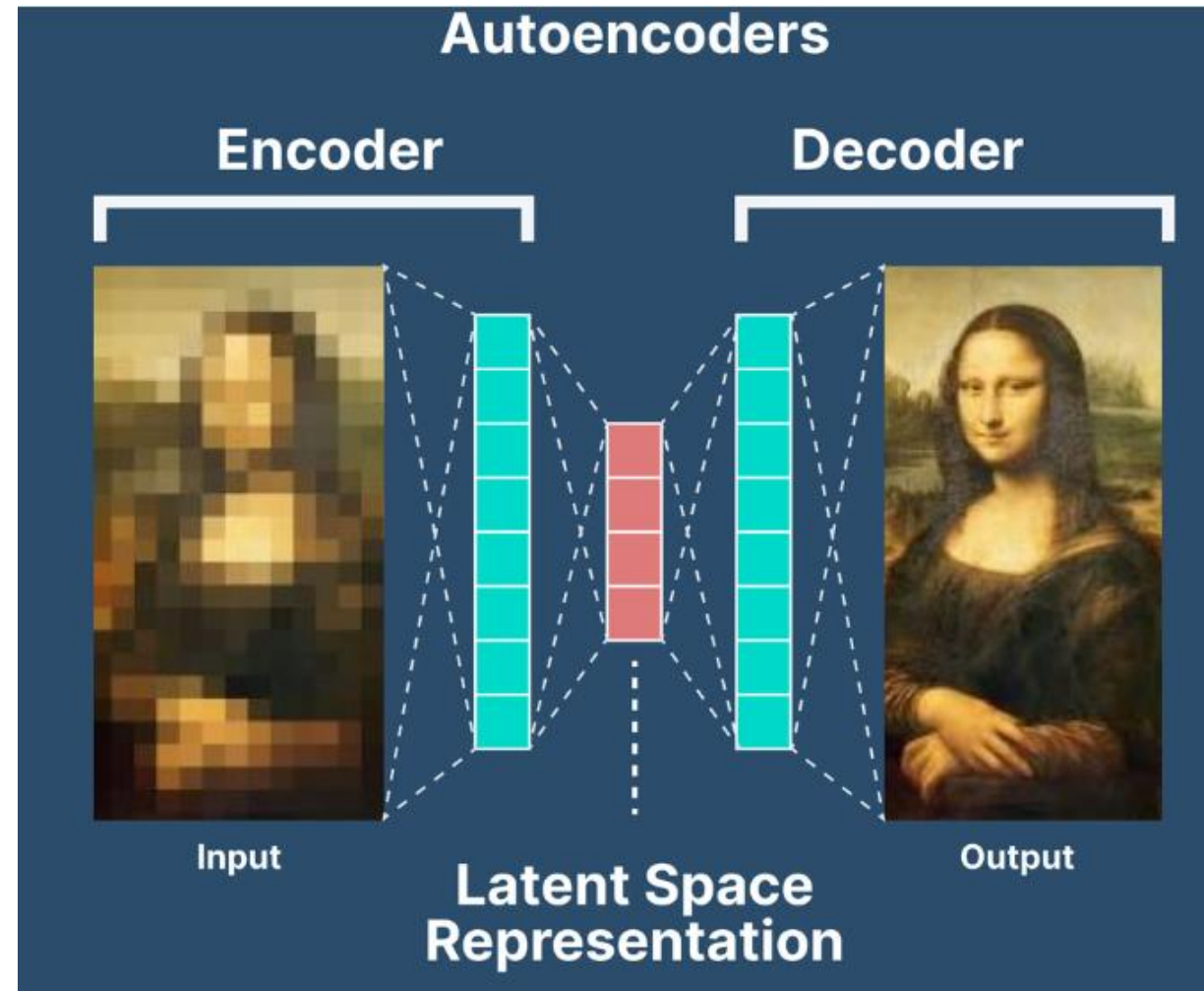
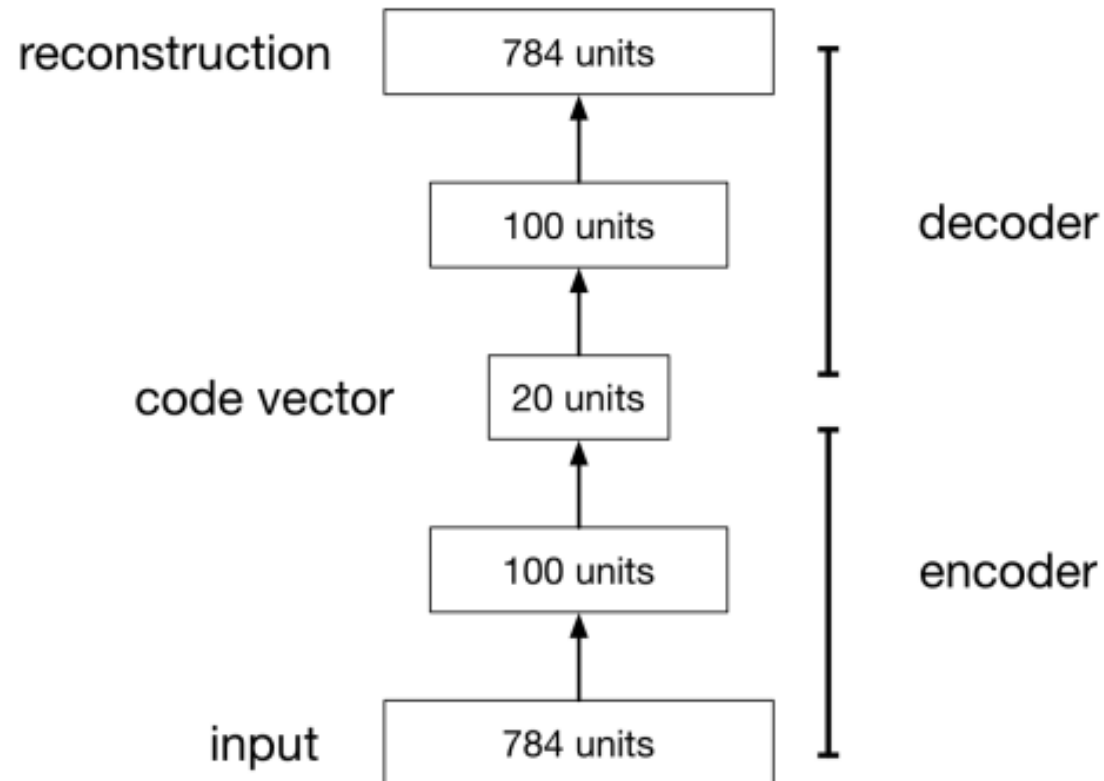


Autoencoders



Autoencoders

- An **autoencoder** is a feed-forward neural net whose job it is to take an input $\mathbf{x}$ and predict $\mathbf{x}$.
- To make this non-trivial, we need to add a **bottleneck layer** whose dimension is much smaller than the input.



Autoencoders

Why autoencoders?

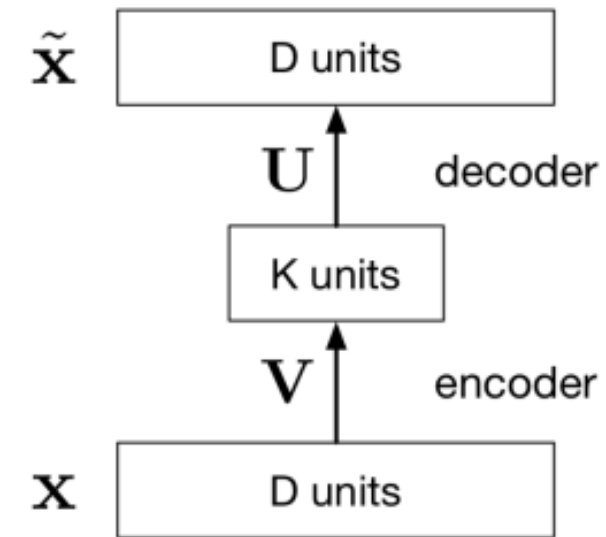
- Map high-dimensional data to two dimensions for visualization
- Compression (i.e. reducing the file size)
- Learn abstract features in an unsupervised way so you can apply them to a supervised task
 - Unlabeled data can be much more plentiful than labeled data

Principal Component Analysis

- The simplest kind of autoencoder has one hidden layer, linear activations, and squared error loss.

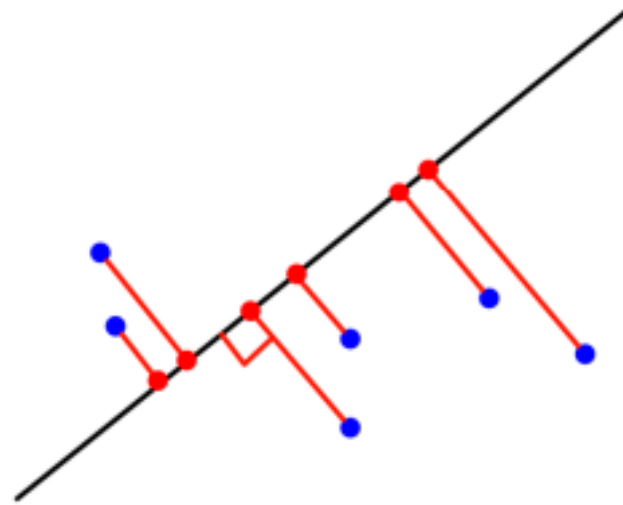
$$\mathcal{L}(\mathbf{x}, \tilde{\mathbf{x}}) = \|\mathbf{x} - \tilde{\mathbf{x}}\|^2$$

- This network computes $\tilde{\mathbf{x}} = \mathbf{U}\mathbf{V}\mathbf{x}$, which is a linear function.
- If $K \geq D$, we can choose $\mathbf{U}$ and $\mathbf{V}$ such that $\mathbf{U}\mathbf{V}$ is the identity. This isn't very interesting.
- But suppose $K < D$:
 - $\mathbf{V}$ maps $\mathbf{x}$ to a K -dimensional space, so it's doing dimensionality reduction.
 - The output must lie in a K -dimensional subspace, namely the column space of $\mathbf{U}$.



Principal Component Analysis

- We just saw that a linear autoencoder has to map D -dimensional inputs to a K -dimensional subspace $\mathcal{S}$.
- Knowing this, what is the best possible mapping it can choose?
 - By definition, the **projection** of $\mathbf{x}$ onto $\mathcal{S}$ is the point in $\mathcal{S}$ which minimizes the distance to $\mathbf{x}$.



Deep Autoencoders

- Deep nonlinear autoencoders learn to project the data, not c subspace, but onto a nonlinear **manifold**
- This manifold is the image of the decoder.
- This is a kind of **nonlinear dimensionality reduction**.

